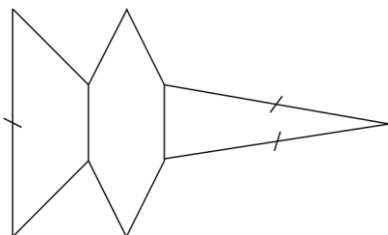




Identifying and naming 2D shapes

Teacher resource

Question 1



Which three figures were used to make this picture?

- ☐ Hexagon, parallelogram, scalene triangle
- ☐ Equilateral triangle, hexagon, rhombus
- ☐ Isosceles triangle, irregular hexagon, trapezium
- ☐ Quadrilateral, regular hexagon, isosceles triangle

Answer: C

Skill: Students accurately name 2D figures.

Background information

Some secondary students may be able to identify and name a two dimensional figure but do not understand that the properties of a class of figures define that class. For example, to show a quadrilateral is a kite, it must have two pairs of adjacent equal sides. This is quite different from the process of identifying a figure as a kite because it 'looks like a kite' and then assuming it must therefore have two pairs of adjacent equal sides because it 'is' a kite.

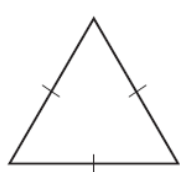
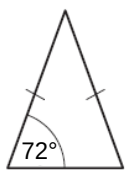
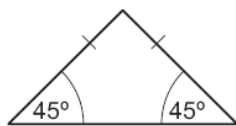
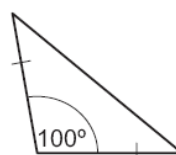
Students will need opportunities to develop an understanding of angle relationships within and the connections between properties of figures. Many students will require physical experiences of constructing figures and then folding, cutting, comparing and re-arranging parts to informally observe relationships between properties before more formal proofs will make sense to them.

Additional questions

1. What changes would need to be made to the picture to make statement D true?
2. What changes would need to be made to the picture to make statement A true?

Question 2

Which one of these is a right-angled isosceles triangle?


☐

☐

☐

☐

Not to scale

Answer: C

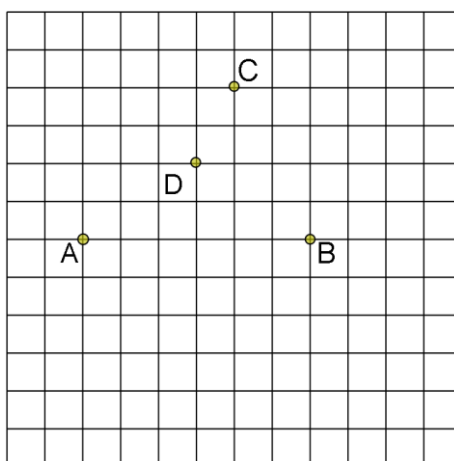
Skill: Students use properties of triangles.

Additional questions

1. What is the most mathematically accurate name for each of triangles A, B and D?
2. Calculate the size of all unmarked angles in the four triangles.

Question 3

Point E can be placed on the grid so that the figure ABCDE has exactly one line of symmetry.



Which of the following is an accurate name for the figure ABCDE?

- ☐ Regular pentagon
- ☐ Concave pentagon
- ☐ Concave regular pentagon
- ☐ Concave isosceles triangle

Answer: B

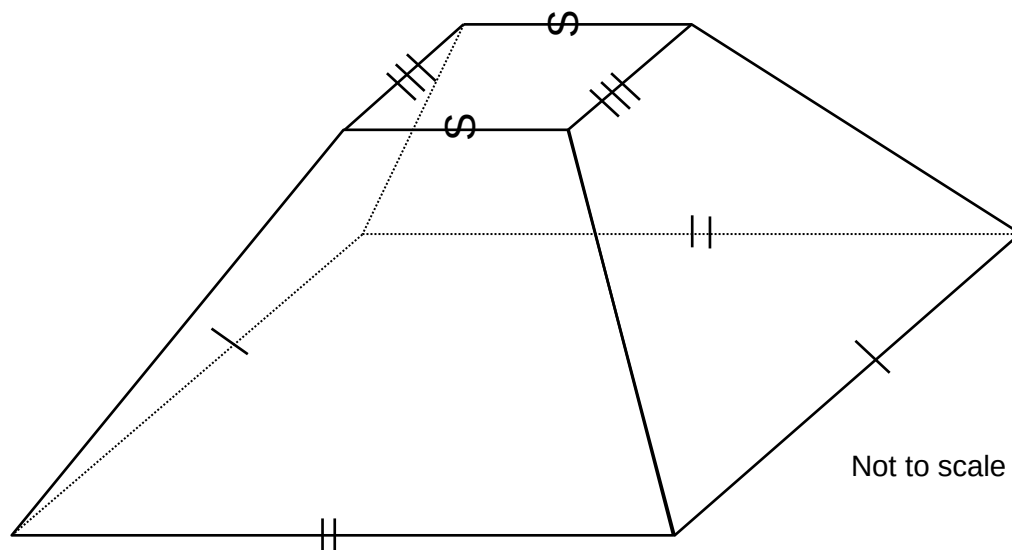
Skill: Students name figures using specific geometric language.

Additional questions

1. Explain why a 'concave regular pentagon' cannot exist.
2. Explain why it is not possible to construct a concave triangle.

Question 4

The diagram below shows a truncated pyramid.



Which of the following best describes the faces of the object?

- ☐ 2 rhombuses, 4 trapeziums
- ☐ 6 parallelograms
- ☐ 2 rhombuses, 2 parallelograms, 2 trapeziums
- ☐ 2 rectangles, 4 trapeziums
- ☐ 2 irregular quadrilaterals, 2 trapeziums, 2 rhombuses

Answer: D

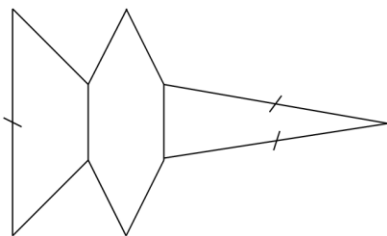
Skill: Students name geometric figures.

Additional questions

1. How many different figures would be used to construct a net of this truncated rectangular pyramid?
2. How many different figures would be required to construct the net of a square based truncated pyramid?

Student activity: Identifying and naming 2D shapes

Question 1

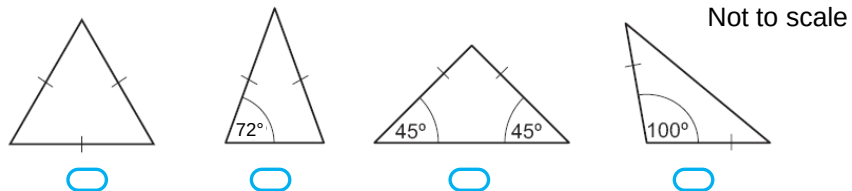


Which three figures were used to make this picture?

- ☐ Hexagon, parallelogram, scalene triangle
- ☐ Equilateral triangle, hexagon, rhombus
- ☐ Isosceles triangle, irregular hexagon, trapezium
- ☐ Quadrilateral, regular hexagon, isosceles triangle

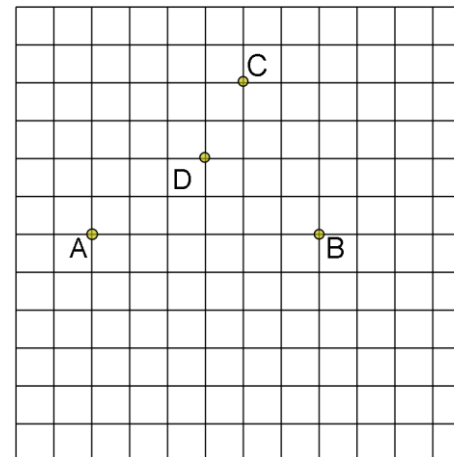
Question 2

Which one of these is a right-angled isosceles triangle?



Question 3

Point E can be placed on the grid so that the figure ABCDE has exactly one line of symmetry.

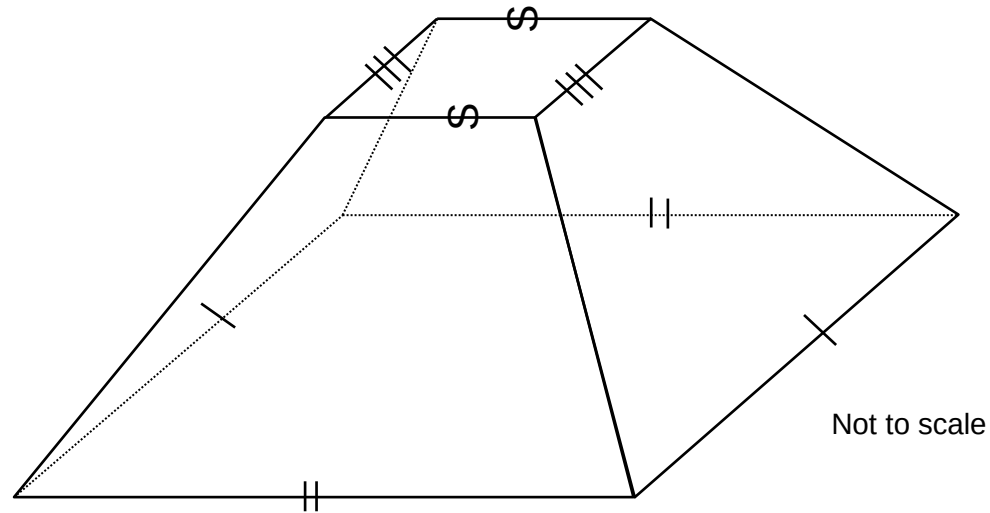


Which of the following is an accurate name for the figure ABCDE?

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- ☐ Concave pentagon
- ☐ Concave regular pentagon
- ☐ Concave isosceles triangle

Question 4

The diagram below shows a truncated pyramid.



Which of the following best describes the faces of the object?

- ☐ 2 rhombuses, 4 trapeziums
- ☐ 6 parallelograms
- ☐ 2 rhombuses, 2 parallelograms, 2 trapeziums
- ☐ 2 rectangles, 4 trapeziums
- ☐ 2 irregular quadrilaterals, 2 trapeziums, 2 rhombuses